

Boas, pythons, and relatives

PHYLUM	Chordata
CLASS	Reptilia
ORDER	Squamata
SUBORDER	Serpentes
SUPERFAMILY	Henophidia
FAMILIES	12
SPECIES	218

Boas and pythons are powerful constricting snakes. They include the world’s largest snakes, among them the anaconda (a boa) and the reticulated, African, and Indian pythons. Boas and pythons do not reproduce in the same way. They also have different distributions: pythons are found in Africa, Asia, and Australasia, while boas occur mainly in

the Americas with a smaller number of species in Africa, Asia, and on some Pacific islands. The relatives of boas and pythons, most of which are also constrictors, include the iridescent sunbeam snakes and the aquatic file snakes.

Anatomy

The snakes in this group are often regarded as being relatively primitive. Their skulls are heavier and their jaws more rigid than those of the more advanced snakes (the colubrids, elapids, and vipers). They have also retained several anatomical features from the limbed animals from which they are descended. These include a back-limb

(pelvic) girdle and, in most species, the remains of back limbs in the form of small claws or spurs. All species have 2 functioning lungs. Several species of boas and pythons have heat-sensitive pits in the scales bordering their mouths, which they use to locate prey in the dark.

Some of the relatives of the boas and pythons are smaller but otherwise outwardly similar. These include wood snakes and Round Island boas. A few species are brightly colored and others, such as sunbeam snakes, have scales that are highly iridescent. File (or wart) snakes are specialized for an aquatic way of life and are effectively helpless on land. Their scales are granular and rough to the touch, an adaptation for gripping and constricting the fishes on which they feed.

Constriction

Although boas, pythons, and their relatives are not the only snakes that kill by constriction (some colubrids also use this method), most of the constrictors are in this group. When the snake has selected its prey, it throws one or more coils around the animal’s body. Each time the victim breathes out, the snake tightens its grip. The victim eventually dies by suffocation, either because it cannot draw breath or because its heart cannot pump blood, rather than by being crushed. Once the prey is dead, the snake loosens its hold and searches for the

CONSERVATION

Several species of boas and pythons, especially the largest ones, have been adversely affected by human activities; these large predators require extensive areas of undisturbed land over which to hunt. In addition, many are relatively conspicuous and sluggish and therefore easy targets for hunters, who kill them for their flesh, their skins, or simply out of prejudice. However, some species perform a useful service to humans by controlling the populations of rodents that are attracted to towns and villages.

head. It swallows this end first, gradually releasing the rest of the body from its coils. Constriction is most effective for killing birds and mammals because, being warm-blooded, they have to breathe relatively frequently.

Reproduction

One of the most significant differences between boas and pythons lies in the way they reproduce. Boas (with the possible exception of one species) bear live young—as do wood snakes, pipe snakes, file snakes, and shield-tailed snakes. Pythons, sunbeam snakes, and Round Island boas lay eggs. Pythons are among the small number of snakes that show parental care. The female coils around the clutch of eggs—which can number up to 100 in large species—throughout the 2–3 month period of incubation, to protect them from predators. One or 2 species of python are unique among snakes in that they can regulate the temperature of their eggs by producing metabolic heat.

FEEDING

A substantial meal can sustain a snake for several weeks or even months. Large constrictors, such as this green anaconda, are capable of killing a variety of animals, including wading birds, deer, young jaguars, and even caimans (as shown here). Despite their great size and weight, anacondas move easily in water and usually ambush their prey in the shallows.

